

Day 110: Classification Model Evaluation

1. Day number

Day 110

2. Topic name

Confusion Matrix, Accuracy, Precision, Recall, and F1 Score

3. Connection

You can now train classification models such as:

```
Logistic Regression
KNN
Naive Bayes
Decision Tree
Random Forest
```

But training a model is only part of machine learning.

Today the question becomes:

| **How do I know whether the model's predictions are actually useful?**

For classification problems, some of the most common evaluation tools are:

```
Confusion Matrix
Accuracy
Precision
Recall
F1 Score
```

4. Important topics

We will use one simple scenario throughout the lesson:

| A model predicts whether an email is **Spam** or **Not Spam**.

Let:

```
Positive class = Spam
Negative class = Not Spam
```

The word **positive** does not mean "good." It simply means the class we have chosen as the positive class.

True Positive — TP

The email really **is Spam**, and the model predicts **Spam**.

Actual: Spam
Predicted: Spam

This is:

True Positive
TP

The model was correct.

True Negative — TN

The email really is **Not Spam**, and the model predicts **Not Spam**.

Actual: Not Spam
Predicted: Not Spam

This is:

True Negative
TN

Again, the model was correct.

False Positive — FP

The email is actually **Not Spam**, but the model predicts **Spam**.

Actual: Not Spam
Predicted: Spam

This is:

False Positive
FP

The model raised a false alarm.

A useful real-world interpretation:

A normal email was incorrectly placed in spam.
--

False Negative — FN

The email really **is Spam**, but the model predicts **Not Spam**.

```
Actual:    Spam
Predicted: Not Spam
```

This is:

```
False Negative
FN
```

The model missed a positive example.

For our scenario:

```
A spam email reached the normal inbox.
```

5. Foundational notes

A classification prediction has two things to compare:

```
Actual label
Predicted label
```

For every example, ask:

```
Was the real class Positive or Negative?
Was the predicted class Positive or Negative?
```

That gives four possibilities:

Actual	Predicted	Result
Positive	Positive	True Positive
Negative	Negative	True Negative
Negative	Positive	False Positive
Positive	Negative	False Negative

A useful memory trick:

First decide whether the prediction was:

```
True  = correct
False = incorrect
```

Then look at what the model predicted:

Positive or Negative

So:

False Positive

means:

| The model predicted Positive, but that prediction was wrong.

6. Confusion matrix

A **confusion matrix** organizes the four counts into a table.

One common layout is:

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

For spam detection:

	Predicted Spam	Predicted Not Spam
Actual Spam	TP	FN
Actual Not Spam	FP	TN

It lets you quickly see not only how many predictions were correct, but also **what kinds of mistakes** the model made.

7. Easy confusion-matrix example

Suppose we tested the model on **10 emails**.

Results:

TP = 4
TN = 4
FP = 1
FN = 1

Confusion matrix:

	Predicted Spam	Predicted Not Spam
Actual Spam	4	1
Actual Not Spam	1	4

Total:

$$4 + 4 + 1 + 1 = 10 \text{ emails}$$

Now we can calculate our main metrics.

8. Accuracy

Accuracy asks:

| Out of all predictions, how many were correct?

Correct predictions are:

$$TP + TN$$

So:

$$\text{Accuracy} = \text{correct predictions} / \text{all predictions}$$

Using our example:

$$\begin{aligned} \text{Correct} &= 4 + 4 = 8 \\ \text{Total} &= 10 \end{aligned}$$

Therefore:

$$\begin{aligned} \text{Accuracy} &= 8 / 10 \\ &= 0.80 \\ &= 80\% \end{aligned}$$

Interpretation:

| The model classified 80% of all emails correctly.

9. Precision

Precision asks:

| When the model predicted Spam, how often was it actually Spam?

Focus only on predictions of the positive class.

Those are:

$$TP + FP$$

Simple formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Using our example:

$$\begin{aligned}\text{TP} &= 4 \\ \text{FP} &= 1\end{aligned}$$

So:

$$\begin{aligned}\text{Precision} &= 4 / (4 + 1) \\ &= 4 / 5 \\ &= 0.80 \\ &= 80\%\end{aligned}$$

Interpretation:

| Of all emails predicted as Spam, 80% really were Spam.

Precision is useful when **false alarms matter**.

For example, high precision is useful if you do not want many legitimate emails incorrectly marked as spam.

10. Recall

Recall asks:

| Out of all the actual Spam emails, how many did the model successfully find?

Focus on actual positives.

Those are:

$$\text{TP} + \text{FN}$$

Simple formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Using our example:

$$\begin{aligned}\text{TP} &= 4 \\ \text{FN} &= 1\end{aligned}$$

So:

$$\begin{aligned}\text{Recall} &= 4 / (4 + 1) \\ &= 4 / 5 \\ &= 0.80 \\ &= 80\%\end{aligned}$$

Interpretation:

| The model detected 80% of all actual Spam emails.

Recall matters when **missing positive cases is costly**.

11. F1 score

Sometimes we care about both:

Precision
and
Recall

The **F1 score** combines them into one number.

A simple formula is:

$$\text{F1} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

In our example:

Precision = 0.80
Recall = 0.80

So:

$$\text{F1} = 0.80$$

or:

80%

At beginner level, remember:

| **F1 is useful when you want a balance between precision and recall.**

You do not need to memorize the formula immediately. Understanding what it represents is more important.

12. All four metrics in one scenario

Using:

TP = 4
TN = 4
FP = 1
FN = 1

we get:

Metric	Beginner question	Result
Accuracy	How many predictions were correct overall?	80%
Precision	When we predicted Spam, how often were we right?	80%
Recall	How many actual Spam emails did we find?	80%
F1	How balanced are precision and recall?	80%

Notice that all four happen to be equal in this very balanced example.

That will **not** always happen.

13. Why accuracy alone can be misleading

Consider a very different dataset:

100 emails total

95 = Not Spam
5 = Spam

Imagine a bad model that predicts:

Not Spam

for **every single email**.

It gets:

95 correct
5 wrong

Accuracy:

$95 / 100 = 95\%$

That sounds excellent.

But the model found:

0 out of 5 Spam emails

Its recall for Spam would be:

0%

So although accuracy is **95%**, the model is useless if our main goal is to catch spam.

This happens with **imbalanced datasets**, where one class is much more common than another.

Main lesson:

| Never judge every classification model using accuracy alone.

Look at the confusion matrix and other metrics too.

14. Problem statement

Suppose these are the actual and predicted labels for eight emails:

Email	Actual	Predicted
1	Spam	Spam
2	Spam	Spam
3	Spam	Not Spam
4	Not Spam	Not Spam
5	Not Spam	Spam
6	Not Spam	Not Spam
7	Spam	Spam
8	Not Spam	Not Spam

Treat:

Spam = Positive
Not Spam = Negative

Your tasks are:

1. Count the **True Positives**.
2. Count the **True Negatives**.
3. Count the **False Positives**.
4. Count the **False Negatives**.
5. Build the confusion matrix.

6. Calculate accuracy.
 7. Calculate precision.
 8. Calculate recall.
 9. Calculate the F1 score.
 10. Explain each result in one simple sentence.
-

15. Concepts used

This exercise combines:

Actual label
Predicted label
Positive class
Negative class
TP
TN
FP
FN
Confusion matrix
Accuracy
Precision
Recall
F1 score

The most important starting point is choosing which class is considered **positive**.

Here:

Positive = Spam

Without knowing that, TP, FP, FN, and TN can become confusing.

16. Thought process

Do not calculate the metrics immediately.

First compare each row individually.

For example:

Actual Spam
Predicted Spam

means:

True Positive

Then:

Actual Spam Predicted Not Spam

means:

False Negative

Continue until every example has been assigned to one of:

TP TN FP FN

Only then calculate the metrics.

This makes mistakes much less likely.

17. Beginner-friendly calculation steps

Follow this order.

Step 1 — Choose the positive class

Positive = Spam Negative = Not Spam
--

Step 2 — Count TP

Find rows where:

Actual = Spam Predicted = Spam

Step 3 — Count TN

Find rows where:

Actual = Not Spam Predicted = Not Spam

Step 4 — Count FP

Find rows where:

Actual = Not Spam
Predicted = Spam

Step 5 — Count FN

Find rows where:

Actual = Spam
Predicted = Not Spam

Step 6 — Build the confusion matrix

	Predicted	
	+	-
Actual +	TP	FN
Actual -	FP	TN

Step 7 — Accuracy

Accuracy =
 $(TP + TN) / \text{Total}$

Step 8 — Precision

Precision =
 $TP / (TP + FP)$

Step 9 — Recall

Recall =
 $TP / (TP + FN)$

Step 10 — F1

Use your precision and recall:

F1 =
$$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

18. Easy edge cases

No predicted positives

Suppose the model predicts every example as:

Negative

Then:

$$TP + FP = 0$$

The usual precision calculation would involve division by zero.

Libraries such as scikit-learn have rules for handling this situation.

At beginner level, simply recognize:

Precision cannot be meaningfully calculated in the normal way if the model never predicts the positive class.

No actual positives

Suppose your test set contains no Spam emails.

Then:

$$TP + FN = 0$$

Recall becomes problematic because there were no actual positive examples to find.

This is another reason test data should represent the classes you care about.

Perfect model

If:

$$FP = 0$$

$$FN = 0$$

then every prediction is correct.

You would normally have:

$$\text{Accuracy} = 1$$

$$\text{Precision} = 1$$

$$\text{Recall} = 1$$

$$F1 = 1$$

which means:

100%

Highly imbalanced data

If one class greatly outnumbers another, accuracy may appear high even when the minority class is handled badly.

Always inspect more than one metric.

19. Expected interpretation

After calculating your exercise, your explanation should sound something like:

Accuracy:
The model correctly classified ____% of all emails.

Precision:
Of all emails predicted as Spam,
____% were really Spam.

Recall:
Of all actual Spam emails,
the model successfully detected ____%.

F1:
The model's balance between precision
and recall is ____.

This interpretation is often more useful than simply writing four decimal values.

For example:

Precision = 0.75

is easier to understand as:

| 75% of the examples predicted as positive were actually positive.

20. Hint only

For the eight-email exercise, first create four counters:

TP = ?
TN = ?
FP = ?
FN = ?

Go through the rows one by one.

Remember:

Actual Spam + Predicted Spam

→ TP

Actual Not Spam + Predicted Not Spam

→ TN

Actual Not Spam + Predicted Spam

→ FP

Actual Spam + Predicted Not Spam

→ FN

Then use:

Accuracy → correct overall

Precision → correctness of positive predictions

Recall → how many real positives were found

F1 → balance of precision and recall

The key idea from **Day 110** is:

A confusion matrix shows what kinds of predictions a classifier got right and wrong, while accuracy, precision, recall, and F1 summarize different aspects of that performance.